DGMS 2019 GEOLOGY • QUESTION 36

Conoscopic Optical Mineralogy: Uniaxial Flash Figures

Topic: Optical Mineralogy & Conoscopic Microscopy | Primary Reference: Dexter Perkins (Mineralogy, 3rd ed., Ch. 5)

Question Statement

Q.36. The interference figure produced by a uniaxial crystal cut parallel to its optic axis is known as

(a) Dichroic figure
(b) Flash figure
(c) Brewster's figure
(d) Uniaxial indicatrix

a Dichroic figure	b **Flash figure (Correct Answer)**
c Brewster's figure	d Uniaxial indicatrix

✓ Correct Answer: (b) Flash figure [When a uniaxial crystal is cut parallel to the optic axis, broad fuzzy isogyres flash across the field of view and leave within a few degrees of stage rotation].

Textbook Evidence: Dexter Perkins (Chapter 5)

of 45°, the isogyres are at maximum separation; they come back together to reform the cross with further stage rotation. The maximum amount of isogyre (Figure 24) of different colors. If the retardation of the crystal is great, isochromes circle the melatopes; none are apparent in Plate 4.10. Interference colors

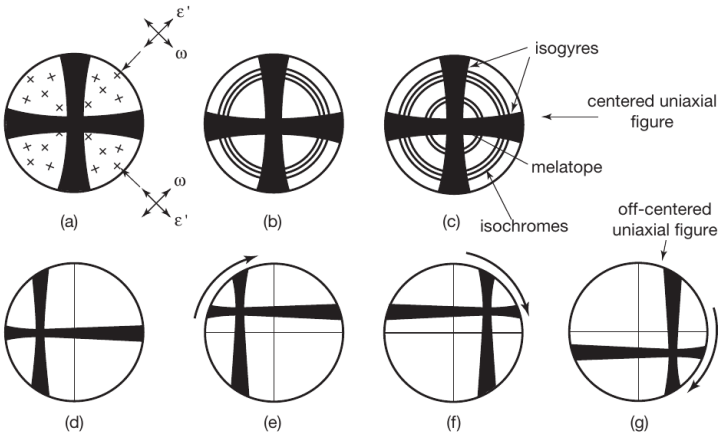


FIGURE 19 Uniaxial interference figures: In (a), (b), and (c) the figure is centered; in (d), (e), (f), and (g) it is off center but the cross is visible. In (a), the small crosses show the vibration directions of ε' and ω; ε' always vibrates radially and ω vibrates perpendicular to ε'. In (b), one set of isochromes is shown, while in (c) there are two sets due to higher birefringence. Figures (d) through (g) show how the melatope (the center of the cross) of an off-center figure precesses as the microscope stage is rotated. See also Plates 4.8 and 4.9.

Figure 5.19: Uniaxial Interference Figures Under Conoscopic Observation (Dexter Perkins, *Mineralogy*, 3rd ed., p. 127). (a–c) Centered optic axis (OA) figures formed when the crystal is cut perpendicular to the optic axis (c-axis), producing an invariant black cross (melatope at center) with concentric isochromes. (d–g) Off-centered OA figures where the melatope precesses. When a crystal is instead cut **parallel to the optic axis**, the optic axis lies in the plane of the section; at extinction, broad diffuse isogyres fill the field, but upon rotating the stage by less than 5°, they split and flash out of view—forming a classic **Flash Figure**.

> *"When a uniaxial mineral is oriented with its optic axis parallel to the stage (cut parallel to the optic axis), it exhibits maximum birefringence under orthoscopic illumination. Conoscopically, when the optic axis is parallel to either polarizer, a very broad, fuzzy dark cross covers nearly the whole field of view. As soon as the stage is rotated by just 2° to 5°, the isogyres rapidly split into hyperbolas that exit the field in opposite quadrants. Because the cross appears and disappears so quickly upon slight rotation, this is termed a* **flash figure**.*"*
>
> **— Dexter Perkins, *Mineralogy*, Chapter 5 (Optical Mineralogy), pp. 126–128**

Comprehensive Option Analysis

(b) Flash figure (Optic Normal Figure) **[CORRECT]**

Optical Mechanism: When a uniaxial mineral (such as quartz, calcite, or apatite) is cut parallel to its c-crystallographic axis (the optic axis), the optic axis lies

horizontally on the microscope stage. Under convergent polarized light (Bertrand lens inserted):

- At the extinction position (optic axis parallel to polarizer or analyzer), light traversing the crystal experiences zero or 90° vibration match, resulting in broad, diffuse, shadowy extinction bands that cover the central field of view.
- Upon slight rotation of the microscope stage (typically < 5°), the retardation increases rapidly across the field, and the fuzzy isogyres break apart and flash out of the field of view into the quadrants of the optic axis.
- This diagnostic phenomenon gives it the universally accepted optical mineralogy name: **Flash Figure**. Grains exhibiting a flash figure always display the *maximum interference color (maximum birefringence)* for that mineral in thin section.

(a) Dichroic figure **[INCORRECT]**

Dichroism (pleochroism in uniaxial minerals) is the change in absorption color of a mineral when rotated in plane-polarized light (PPL) due to differential absorption along ω and ϵ vibrations. There is no such conoscopic interference feature known as a "dichroic figure".

(c) Brewster's figure **[INCORRECT]**

Named after Sir David Brewster, Brewster's fringes or figures occur in interferometry and photoelasticity of stressed isotropic media (birefringence induced under mechanical stress). It is not a standard crystallographic interference figure of unstrained minerals.

(d) Uniaxial indicatrix **[INCORRECT]**

The *uniaxial indicatrix* is an abstract three-dimensional geometric ellipsoid of revolution used to represent the refractive indices and vibration directions of a crystal mathematically (n_ω and n_ϵ). It is a theoretical geometric construct, NOT an observed optical interference figure seen down the microscope.

Uniaxial Interference Figures by Grain Cut Orientation

Cut Orientation	Orthoscopic Appearance	Conoscopic Figure Name	Figure Appearance on Stage Rotation
Perpendicular to Optic Axis ($\perp c$-axis)	Zero birefringence; stays dark (extinct) during 360° rotation	**Centered Optic Axis (OA) Figure**	Sharp black cross whose center (melatope) remains fixed at the crosshairs; concentric circular isochromes.
Inclined to Optic Axis	Low to moderate interference colors; extinct every 90°	**Off-Centered Optic Axis Figure**	Black cross whose melatope precesses in a circle around the center of the field as the stage is rotated.
Parallel to Optic Axis ($\parallel c$-axis)	**Maximum interference colors (highest birefringence)**	**Flash Figure (Optic Normal Figure)**	**Broad, fuzzy isogyres cover the field at extinction; split and rapidly flash out of view within 5° rotation.**

Key Takeaway & Exam Strategy

Essential Diagnostic Rules for Optical Mineralogy:

- **Cut $\perp$ Optic Axis:** Minimum birefringence (isotropic behavior in thin section) $\rightarrow$ **Centered Optic Axis Figure** (stable centered black cross).
- **Cut $\parallel$ Optic Axis:** Maximum birefringence $\rightarrow$ **Flash Figure** (fleeting broad isogyres that flash out upon minimal rotation).
- Remember: A flash figure confirms the mineral is anisotropic and oriented with its optic axis horizontal, but cannot easily be used to determine optic sign.